VigilODE polyalgorithm extensions: Fused polynomial actions, finite-step error constants, and causal certification

VigilODE research supplement

(Dated: 28 October 2026)

We investigate the six remaining development tasks following PR92, reusing the existing conditioned resolvent, joint polynomial action, signed-adjoint certificate, causal cache, and homotopy implementation. Three parallel theory lanes produce seventeen concrete alternatives. The leading constructions are a fused vector-coefficient Clenshaw action, a cubic midpoint defect bound with constant 1/12 and rational decay accumulation, variable-step two-channel transport, and value-only continuation margins. We distinguish established results, new project-specific derivations, finite exact-arithmetic checks, and unimplemented floating-point obligations. Three programs, published before execution, pass their declared exact-rational checks. On the stated targets, fusion changes operator products from $5m$ to m, and the cubic planning example changes its minimum dyadic grid from 512 to 256. These are operation and planning results, not measured solver speedups. Runtime defaults, Rust source, prior negative evidence, and unresolved deployment qualification are preserved.

I. QUESTION, SOURCE IDENTITY, AND USEFUL REUSE

The inspected runtime baseline is PR92, commit `7eb2dd0ffda39afb2871d76700a720986ff09fe0`.

The initial remote survey found no later pull request. Task labels PR-N01 through PR-N06 denote planned development tasks, not GitHub pull-request numbers. This supplement executes the requested literature and constructive-exploration stages R2/R4 of PhysMath Research Loop 1.3.0, with supporting derivations and checks. It does not restart earlier ten-stage campaigns or their correction budgets.

Several proposed optimizations are already present. PR68 added joint actions in Chebyshev and Laguerre bases; PR69 preserved an unsuccessful continuous-scaling experiment. The latest code subsequently added signed output-adjoint Laguerre certification and guarded Leja/Chebyshev/Laguerre routing. The coupled solver already shares one projected factorization across eight stages. Causal finite propagation and a nilpotent error majorant were already proved in PR91. These are baselines to reuse, not new achievements here.

The current guarded polynomial provider is an immutable *dense symmetric* operator. Its mathematical results do not automatically certify an opaque matrix-free callback. Matrix-free, Jacobian-

TABLE I. Existing capabilities and the proposed increment. Source paths are relative to the repository's `crates/` directory.

Current source	Already available	Proposed increment
`rodas5p-core/` polynomial action and router	Coefficient enclosures, complete-total admission, exact cache keys, charged refusals	Fused-only backward execution with its own local-error certificate
`rodas5p-integrators/` conditioned polynomial	Full-space relation, conditioned chart, original residual	Prefer cheap structure; extend by a proved perturbation bound
`vigilode-coupled/` causal and structural modules	Shared factor, prefix reuse, represented-stage residual enclosure	Physical and variable-step transport; carrier-specific endpoint budgets
`vigilode/` session and switcher	Scoped persistent session, same-window work comparison	Retain winning session; conditional future residence with an explicit cost-variation certificate

free, and derivative-free are different properties. A model-declared generator can satisfy all three; a supplied JVP remains derivative-assisted even without a stored matrix.

II. CRITICAL USE OF THE SUPPLIED LITERATURE

Robust numerical algebraic geometry studies parameter uncertainty and exceptional algebraic strata[1]. Searching for nearby parameters can clarify the structure of an uncertain polynomial system, but changing an ODE parameter would change the original integration target. We reuse explicit branch and parameter contracts, without importing parameter regularization as an implicit acceptance rule.

The numerical nonlinear algebra survey distinguishes regular parameter paths, root tracking, and certification[2]. Its Newton-based tracking and Jacobian-based Krawczyk method do not substantiate a derivative-free implementation. A generic complex continuation theorem also does not identify the prescribed real physical branch.

Kunkel–Mehrmann require regularity, derivative-array rank conditions, coordinate choices and reconstruction for inherent ODEs[3]. Their geometric symmetry construction is for linear time-varying DAEs. The paper explicitly notes that it is inapplicable to its nonlinear pendulum example, despite a self-adjoint Jacobian pair. A declared analytic reduction is a useful future provider;

generic RHS-only nonlinear DAE support remains open.

The ETH notes describe Hamiltonian composition and the Runge–Kutta identities for invariants[4]. They motivate distinct structure-preserving routes, while leaving adaptive selection and inexact-stage effects to be analyzed separately.

III. FUSED BACKWARD POLYNOMIAL ACTIONS

For five distinct inputs and an output that requests only their sum, write

$$F_m = \sum_{j=0}^{4} \sum_{k=0}^{m} c_{kj} P_k(X) w_j = \sum_{k=0}^{m} P_k(X) d_k, \qquad d_k = \sum_{j=0}^{4} c_{kj} w_j. \tag{1}$$

The coefficient vectors can be formed on demand. The existing forward implementation uses five vector recurrences for distinct inputs. Equation (1) permits one backward recurrence.

A. Laguerre identity and error transfer

Set $b_{m+1} = b_{m+2} = 0$ and evaluate

$$b_k = d_k + \frac{(2k+1)I - X}{k+1} b_{k+1} - \frac{k+1}{k+2} b_{k+2}, \qquad k = m, \ldots, 0. \tag{2}$$

Then $b_0 = F_m$ for ordinary Laguerre polynomials. If each computed recurrence has additive local defect δ_k, including coefficient-vector formation, operator-action error and rounded arithmetic, the exact identity is

$$\widehat{b}_0 - F_m = \sum_{k=0}^{m} L_k(X) \delta_k. \tag{3}$$

Proof. The transfer of an impulse at position k obeys the Laguerre recurrence, with initial polynomials 1 and $1 - x$. It is therefore $L_k(x)$. Superposition proves the action and perturbation identities, including vector-valued coefficients. If X is self-adjoint in the declared Hilbert norm and $\sigma(X) \subset [0, L]$, the spectral theorem and DLMF 18.14.8[5] imply

$$\left\| \widehat{b}_0 - F_m \right\| \leq e^{L/2} \sum_k \|\delta_k\| . \tag{4}$$

This bound is independent of degree in its transfer factor, but local defects can grow with intermediate vectors. Large L can make the bound unhelpful. It is not valid to apply this factor to defects from the old forward execution; the local defects depend on the evaluation direction.

For Chebyshev polynomials, use

$$b_k = d_k + 2Xb_{k+1} - b_{k+2} \quad (k \geq 1), \qquad F_m = d_0 + Xb_1 - b_2. \tag{5}$$

Telescoping gives the defect transfer $\sum_k T_k(X)\delta_k$. On a self-adjoint represented chart contained in $[-1, 1]$, its bound is $\sum_k \|\delta_k\|$. If rounding expands the interval, use the corresponding $\cosh(k \operatorname{arcosh}(1 + \eta))$ growth bound.

B. Cost and admission

At degree m, the fused-only distinct-input target changes operator products from $5m$ to m. Coefficient-vector assembly still costs $O(5mn)$ scalar operations. The current same-vector input already uses m actions; callers requesting all individual columns cannot use the shortcut. There is no factor-five elapsed-time claim.

The new complete budget must include truncation, coefficient enclosures, source combinations, recurrence defects, normalization, underflow and output assembly. The current signed-adjoint Bernstein envelopes remain the comparator. Their existing resource cap is not removed by an unimplemented bound. A later matrix-free port needs immutable provider identity and a certified action-error enclosure; any extra action or matrix storage used for that certificate belongs in the cost.

IV. RESOLVENTS, HYBRIDS, AND NONNORMAL ALTERNATIVES

A. Exact Chebyshev degree

For self-adjoint positive A, $\sigma(A) \subset [a, b]$, $0 < a < b$, set $c = (a + b)/2$, $d = (b - a)/2$, and

$$p_m(z) = \frac{T_m((c - z)/d)}{T_m(c/d)}, \qquad q_{m-1}(z) = \frac{1 - p_m(z)}{z}. \tag{6}$$

Since $p_m(0) = 1$, q_{m-1} is polynomial. Updating $x_m = x_0 + q_{m-1}(A)r_0$ gives $r_m = p_m(A)r_0$ and

$$\|r_m\| \leq \delta_m \|r_0\|, \qquad \delta_m = \frac{1}{T_m(c/d)} = \frac{2q^m}{1 + q^{2m}}, \tag{7}$$

$$q = \frac{\sqrt{b/a} - 1}{\sqrt{b/a} + 1}, \qquad m_{\min} = \left\lceil \frac{\operatorname{arcosh}(1/\varepsilon)}{\operatorname{arcosh}(c/d)} \right\rceil. \tag{8}$$

The spectral theorem gives the norm bound. Alternation of T_m gives the minimax residual polynomial: a strictly better polynomial would have a difference with m roots between the extrema and an additional root at zero outside the interval. This exceeds its degree.

An outward integer check must validate the chosen degree; rounded logarithms alone are insufficient. For $[a, b] = [1, 9]$, $q = 1/2$ and the exact smallest degree for $\delta_m \leq 10^{-8}$ is 28. If $a = b$, self-adjointness gives $A = aI$, so a scalar solve is the appropriate route.

These methods eliminate orthogonalization and basis storage, but still require dependent operator applications. Compare against CG, the existing shared coupled basis, and specialized direct relations, not only a weak GMRES implementation. Polynomial preconditioners can have useful fixed linearity properties[6], while Laguerre short recurrences trade inner products and storage against approximation behavior[7].

B. Reuse an approximate structural inverse

Let P satisfy $\|D\| \leq \theta < 1$ for $D = I - AP$. Define

$$Q_r = P \sum_{j=0}^{r-1} D^j. \qquad AQ_r = I - D^r, \qquad \|f - AQ_r f\| \leq \theta^r \|f\|. \tag{9}$$

The finite geometric sum proves this identity without requiring $AP = PA$. For $A = A_0 + E$, a sufficient bound is $\theta \leq \|I - A_0 P\| + \|E\| \, \|P\|$. Existing exact exchange, block and quotient inverses can therefore serve as cheap near-structure correction operators. Freeze the polynomial plan within a standard linear outer solver, or use an explicitly flexible method.

C. Nonnormality and numerical range

For $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $A = I + KN$, and $p(z) = 1 - z$, all eigenvalues equal one but $\|p(A)\|_2 = |K|$. Eigenvalue placement alone does not justify the SPD bound. Yet $A^{-1} = I - KN$ is exact: the same example favors the existing declared low-degree relation route.

A verified numerical range permits the established constant [8]

$$\|g(A)\|_M \leq (1 + \sqrt{2}) \sup_{z \in W_M(A)} |g(z)|. \tag{10}$$

If $C = (cI - A)/d$ has numerical range inside a filled Bernstein ellipse of parameter R, with $1 \leq R < \zeta = \xi + \sqrt{\xi^2 - 1}$, $\xi = c/d > 1$, then

$$\|p_m(A)\|_M \leq (1 + \sqrt{2}) \frac{R^m + R^{-m}}{\zeta^m + \zeta^{-m}}. \tag{11}$$

The ellipse identity for T_m and the maximum principle prove this bound. It is a first Chebyshev/Faber branch requiring no general Faber engine. General Leja/Faber choices need scalar and

operator-domain error bounds [9, 10]. Matrix-function experiments show that the best backend depends on its integrator and problem[11]; a universal ordering is not inferred.

V. A FINITE-STEP CUBIC CONSTANT AND A SMALLER JUSTIFIED GRID

Consider $z' = -\phi(z)$, $\phi(z) = kz + gz^3$, $k > 0$, $g \geq 0$, $z(0) = Y \geq 0$. Let $L(u) = k + 3gu^2$, $D_h = 1 + hk/2$, and $F_u(K) = K + h\phi(u + K/2)$. For the exact endpoints $u = z(t)$, $v = z(t + h)$,

$$|F_u(v - u)| \leq \frac{L(u)^2\phi(u)h^3}{12}, \qquad |M_h(u) - v| \leq \frac{L(u)^2\phi(u)h^3}{12D_h}. \tag{12}$$

These are finite-step inequalities for every $h > 0$.

Proof. The exact trajectory is decreasing and convex, with $0 \leq z'' \leq L(u)\phi(u)$. If $\ell(s)$ is its endpoint chord and $m = (u + v)/2$, then $0 \leq \ell(s) - z(t + s) \leq L(u)\phi(u)s(h - s)/2$. Define nonnegative quantities

$$A = \int_0^h [\phi(\ell(s)) - \phi(z(t + s))], ds, \tag{13}$$

$$B = \int_0^h \phi(\ell(s)), ds - h\phi(m) = \frac{ghm(u - v)^2}{4}. \tag{14}$$

The residual is $A - B$. Integrating the chord kernel gives $A \leq L(u)^2\phi(u)h^3/12$. Since $u - v \leq h\phi(u)$ and

$$L(u)^2 - 3gu\phi(u) = k^2 + 3kgu^2 + 6g^2u^4 \geq 0, \tag{15}$$

the same upper bound holds for B. Use $|A - B| \leq \max(A, B)$. Finally $F_u'(K) \geq D_h$ on the real line proves the endpoint bound. Derivatives enter this analytic proof, not the runtime callback contract.

The midpoint map is globally nonexpansive because

$$M_h'(x) = \frac{1 - (h/2)\phi'(m)}{1 + (h/2)\phi'(m)}, \qquad |M_h'(x)| \leq 1. \tag{16}$$

Thus numerical error is transported separately from the exact trajectory used in the local truncation term. Since $z_j \leq Ye^{-kjh} \leq Yq^j$, $q = (1 + kh)^{-1}$, and the polynomial $L(x)^2\phi(x)$ has nonnegative coefficients, a cumulative bound is

$$\mathcal{T}_N = \frac{h^3}{12D_h}\left[k^3YS_N(q) + 7k^2gY^3S_N(q^3) + 15kg^2Y^5S_N(q^5) + 9g^3Y^7S_N(q^7)\right], \tag{17}$$

where $S_N(r) = \sum_{j<N} r^j$. For dyadic N, the updates $(p, s) \mapsto (p^2, s(1 + p))$, starting with $(r, 1)$, compute this sum with $O(\log N)$ positive operations. No transcendental library is needed.

TABLE II. Registered exact-rational planning example: $k = 1$, $g = 1/2$, $Y = T = 1$, $\varepsilon = 10^{-5}$. Displayed decimals are diagnostic; admission comparisons were exact.

Bound	Truncation bound at $N = 256$	Minimum dyadic N
Inherited constant 1/8	$1.7846536915\,10^{-5}$	512
Constant 1/12 only	$1.1897691277\,10^{-5}$	512
1/12 and rational decay	$3.3853071299\,10^{-6}$	256

The remaining budget must cover root error, update rounding and clock displacement through the complete prefix gate. An admissible residual threshold alone does not pay all these terms. The inherited Picard fallback also retains its contraction gate. A separate bracketed monotone backend could relax that backend-specific restriction, but must not silently call an unqualified noncontractive fallback.

VI. STRUCTURAL AND CAUSAL ERROR TRANSPORT

For $P = 11^T/n$, $Q = I - P$, $J = -dQ - (d - \kappa)P$, $d \geq \kappa \geq 0$, a discrete step has $S_k = \sigma_{k,t}Q + \sigma_{k,c}P$. Fixed projectors permit

$$\prod_{\ell>k} S_\ell = p_{k,t}Q + p_{k,c}P, \qquad p_{k,r} = \prod_{\ell>k} \sigma_{\ell,r}. \tag{18}$$

At each h_k, the existing scalar stage matrices $L_k(r)$ give adjoints $w_{k,r} = L_k(r)^{-T}\beta$. A full-stage residual $\rho_{k,i}$ contributes $(w_{k,t,i}Q + w_{k,c,i}P)\rho_{k,i}$ to the endpoint. The exact induced infinity norm of a two-channel block is

$$\nu_n(t, c) = \left|t + \frac{c - t}{n}\right| + (n - 1)\left|\frac{c - t}{n}\right|. \tag{19}$$

Unrolling the error recurrence and applying this norm gives the variable-step residual transport, with sharp independent-block row sums. Temporal defects, endpoint arithmetic and clock discrepancy remain separate terms. Products over an accepted grid are an a posteriori statement, not a certificate for an unknown future adaptive grid.

For the declared affine model $y' = b + Jy$, the exact flow is

$$F_\Delta(y) = e^{-d\Delta}Qy + e^{-(d-\kappa)\Delta}Py \tag{20}$$

$$+ \Delta\varphi_1(-d\Delta)Qb + \Delta\varphi_1(-(d - \kappa)\Delta)Pb. \tag{21}$$

If $\zeta_k = \widehat{y}_{k+1} - F_{\Delta_k}(\widehat{y}_k)$ is enclosed,

$$E_{k+1} \le e^{-(d-\kappa)\Delta_k} E_k + \|\zeta_k\|_\infty . \tag{22}$$

The exponential has nonnegative entries and row sum $e^{-(d-\kappa)t}$, proving the scalar factor. This also suggests a direct flow route when it is cheaper than executing and certifying eight stages. Validated exponential/φ arithmetic is still required.

For the general raw-U causal equations,

$$WU_i = h\gamma f\left(y + \sum_{j<i} a_{ij}U_j\right) + \gamma \sum_{j<i} c_{ij}U_j, \qquad W = I - h\gamma J_0, \tag{23}$$

let $\left\|W^{-1}\right\| \le \omega$ and let L_i bound the RHS Lipschitz constant on the relevant stage domains. Then

$$e \le \omega \sum_{\ell=0}^{s-1} B^\ell \rho, \qquad B_{ij} = \omega(h\gamma L_i|a_{ij}| + |\gamma c_{ij}|), \quad j < i. \tag{24}$$

Subtract the stage equations to obtain $e \le \omega\rho + Be$; strict lower triangularity gives $B^s = 0$. This instantiates the already-proved causal majorant. It does not need a global contraction factor below one.

For a carrier-aware affine stage matrix with inverse block bounds D and nonlinear remainder majorant N, the sharper form is $e \le \sum_{\ell<s}(DN)^\ell D\rho$. Endpoint weights convert this to a state error budget. A changed carrier's temporal or stiff-uniform order remains a separate question.

VII. CONTINUATION, SESSION REUSE AND TEMPORAL RESIDENCE

For $G_\lambda = (1-\lambda)a + \lambda G$ on a closed ball $\overline{B}(c, r)$, suppose G is q-Lipschitz and certified quantities satisfy $e \ge \|G_\lambda(c) - c\|$, $b \ge \|G(c) - a\|$. The next station $\mu = \lambda + \delta \le 1$ leaves the ball invariant if

$$(\lambda + \delta)qr + e + \delta b \le r. \tag{25}$$

This follows by adding and subtracting $G_\mu(c)$ and $G_\lambda(c)$. When the denominator is nonzero and the margin positive,

$$\delta \le \frac{(1 - \lambda q)r - e}{qr + b}, \qquad \mu q < 1 \tag{26}$$

provides a value-only Banach station and residual-to-root constant $1/(1 - \mu q)$. The strict contraction check is separate. If $q \ge 1$, this does not establish the endpoint $\lambda = 1$. If the endpoint is already contractive, intermediate stations may add unnecessary cost.

For a monotone scalar residual, a certified bracket and $F(v)-F(u) \geq m(v-u)$ give uniqueness and error $|F(x)|/m$ without runtime derivatives. Small residual alone cannot choose between the two roots of $x^2 - p = 0$.

Session reuse has a different role. Preserving every state component read by a deterministic integrator, its callback bindings, output landings and arithmetic order implies pause/resume trace equality by induction. This requires FSAL validity, controller history, buffers, cumulative budgets and epochs; a (t, y, h) snapshot is insufficient. The first candidate is to continue a winning probe inside the existing scoped session rather than introducing a self-referential owned callback design.

Finally, same-window cost evidence does not establish future residence. Under an explicit future variation certificate $C_B(0)-C_A(0) \geq d > 0$ and $|(C_B-C_A)(j)-(C_B-C_A)(j-1)| \leq V$, telescoping gives

$$C_B(j) - C_A(j) \geq d - jV, \qquad \sum_{j=1}^{m}(C_B(j) - C_A(j)) \geq md - \frac{Vm(m+1)}{2}. \tag{27}$$

Require $mV < d$ and the latter bound greater than switching/calibration cost S. For continuous rates with Lipschitz constant $L > 0$, the corresponding bound is $dH - LH^2/2$ for $H \leq d/L$. If $L = 0$, all $H \geq 0$ preserve the advantage and strict payback is $dH > S$. Smooth ODE coefficients do not supply such a work certificate: adaptive acceptance thresholds can cause discrete jumps. Trace-margin certification and its cost remain a separate research task.

VIII. GEOMETRIC ALTERNATIVES AND TWO EXACT COUNTEREXAMPLES

Declared separable Hamiltonians admit force-only splitting without Jacobian/JVP/Newton evaluations. For oscillator Verlet,

$$M(h) = \begin{pmatrix} 1 - z^2/2 & h \\ -h\omega^2(1 - z^2/4) & 1 - z^2/2 \end{pmatrix}, \qquad z = h\omega. \tag{28}$$

The determinant is one, and the fixed-step power-bounded interval is $0 < h\omega < 2$. The endpoint has a nontrivial Jordan block. Yet $\omega = 1$, $h_1 = 1$, $h_2 = 19/10$ obey the individual interval and satisfy

$$\det(M(h_2)M(h_1)) = 1, \qquad \operatorname{tr}(M(h_2)M(h_1)) = -9661/4000 < -2. \tag{29}$$

The periodic product is unstable. Moreover, the exact free-particle flow with state-dependent $h(q, p) = \epsilon q$ has determinant $1 + \epsilon p$, so state-dependent selection can destroy symplecticity itself.

An inexact RK stage can nevertheless have a useful invariant-drift certificate. Let $I(y) = y^T C y$, $C = C^T$, and $y^T C f(y) = 0$. For $F_i = f(Y_i) + \eta_i$, stage defects $\delta_i = Y_i - y - h\sum_j a_{ij} F_j$, and $z = y + h\sum_i b_i F_i$, direct quadratic expansion gives

$$\begin{aligned} I(z) - I(y) = & -h^2 \sum_{ij} (b_i a_{ij} + b_j a_{ji} - b_i b_j) F_i^T C F_j \\ & - 2h \sum_i b_i \delta_i^T C F_i + 2h \sum_i b_i Y_i^T C \eta_i. \end{aligned} \tag{30}$$

For a symplectic RK tableau the first term vanishes. Endpoint error r adds $2z^T Cr + r^T Cr$. This bounds invariant drift; it does not make an inexact input-output map exactly symplectic.

Other concrete branches in the catalog are an exact small invariant block plus a polynomial complement, predetermined affine prefix scans, and a declared regular DAE reduction provider. They retain their own structural and reconstruction hypotheses.

IX. VERIFICATION, DEVELOPMENT PRIORITIES AND REMAINING OBLIGATIONS

The three checker programs and their protocol were published before execution: `866668348f840bd43558859d2795576e292b6ca7`.

All passed. Polynomial checks cover degrees 0–8 and all impulse positions, the exact degree-28 example and a Jordan counterexample. Homotopy checks include 108 cubic algebra fixtures, exact grid decisions, continuation-ball and branch controls, and conditional residence. Four geometry checks compare scalar adjoints with independently assembled full block matrices, verify a nonlinear causal majorant with norm greater than one, check invariant drift, and reproduce the Verlet counterexamples.

These checks do not replace the written proofs. They do not validate new binary64 kernels or measure native performance. Earlier exploratory CAS checks are explicitly identified; the later registration does not apply retroactively to them. One bounded cross-lane review corrected the explicit $L = 0$ wording in the continuous residence statement. No checker source or runtime arithmetic changed after the registered run.

The refreshed machine-readable plan preserves 23 inherited nodes, 34 edges, eight native suites, seven unresolved qualification tasks and existing correction budgets. Seventeen candidate records refine the six tasks rather than creating seventeen new mandatory PRs. The deployment sequence remains release qualification, matched-accuracy hardware measurements, and actual client quali-

TABLE III. Six task contracts. All native implementations remain pending.

Task	First implementation candidate	Remaining theoretical/validation bridge
N01	Fused-only backward Clenshaw; retain current coefficient/admission code	New local-defect enclosure, complete cost, matrix-free authority; nonnormal witness
N02	1/12 rational-decay outward planner	Prefix/update/clock budget, minimum-grid and range-boundary oracles
N03	Variable-step two-channel transport; direct-flow comparator	Physical defect enclosure and charged adaptive consumer
N04	Retain winning scoped session and workspace capacity	Full trace identity, generic domain and branch certificate
N05	Raw-U residual-to-endpoint budget and work/span comparison	Nonlinear closure, carrier order, domain and stiff-uniform accuracy
N06	Distinguish observed cost from certified variation	Counterfactual future trace/variation certificate and its acquisition cost

fication. Earlier Givens, fair-AB, SUNDIALS, lint/API, negative control and authority-reference issues are not declared resolved here.

The polyalgorithm first filters candidates by target, structure, derivative permission and physical certificate. Only then does it compare total setup, actions, memory, verification, retries and failed work. Hardware service times are measured quantities; exact numerical constants cannot determine a universal fastest policy. Theory can fix the admissible region and error budget while local evidence decides the economic choice within that region.

Complete proofs, candidate contracts, exact outputs, source-bound receipts, the Korean report and local Codex handoff are stored in `research/polyalgorithm_alternatives_20261028/`. The unchanged PR92 evidence retains its original scope. This supplement supplies a concrete implementation basis without claiming a general nonlinear closure theorem or a new production speedup.

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